COOLING
ENVIRONMENTS

Cooling Environment Project Overview and Update

Steve Mills

Technical Lead - META

OCP PLATINUM Member

Cooling Environments Project Structure

Steering Committee Representative

Steve Mills
Meta

Project Co-Leads

Sean Sivapalan
Nvidia

John Fernandes
Meta

Door Heat Exchanger

Jabari George
(JM Gross Eng)

Martin Doerrich
(Rittal)

CDU

John Menoche

Pardeep Shahi

Hussam Kabbani

Brian Burbank

Heat Reuse

Jaime Comella

Bharath Ramakrishnan

Immersion

Rolf Brink
(Promersion)

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Cold Plate

Berhanu Wondimu
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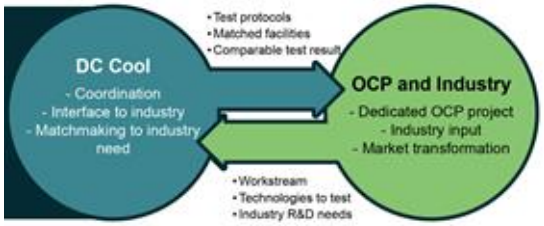
Project Co-Leads	Sean Sivapalan Nvidia	John Fernandes Meta

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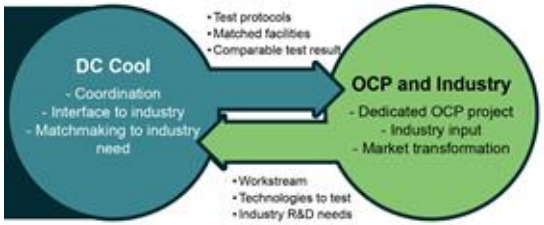
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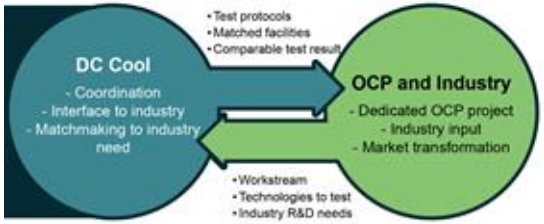




Plate Heat Exchanger
Base Specification

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John Fernandes
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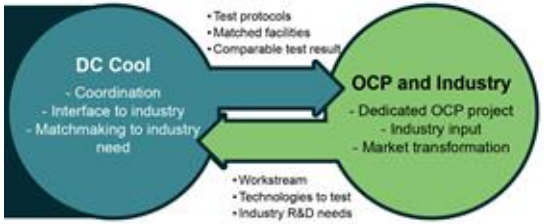




Plate Heat Exchanger
Base Specification



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Thermal Boundaries in AI Infrastructure

Project Co-Leads	Sean Sivapalan Nvidia	John Fernandes Meta

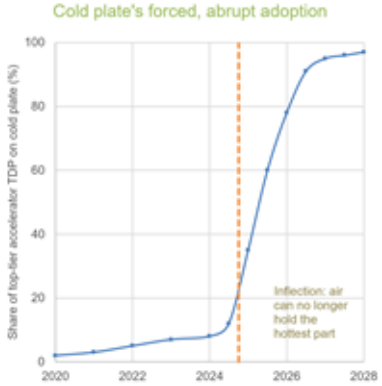
$$\dot{Q} = \dot{m} \cdot c_p \cdot \Delta T$$

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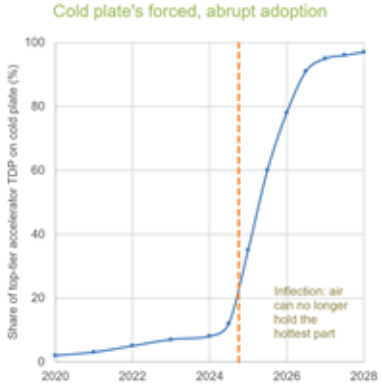


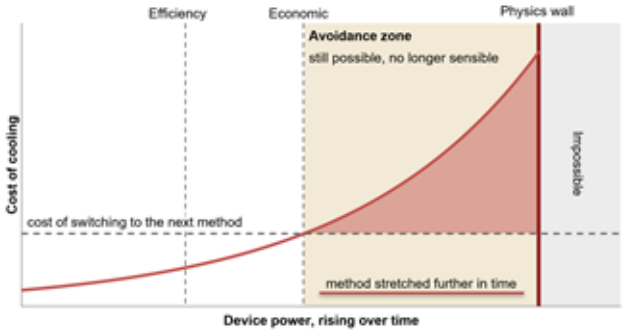
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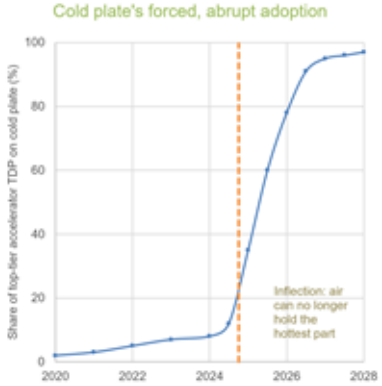


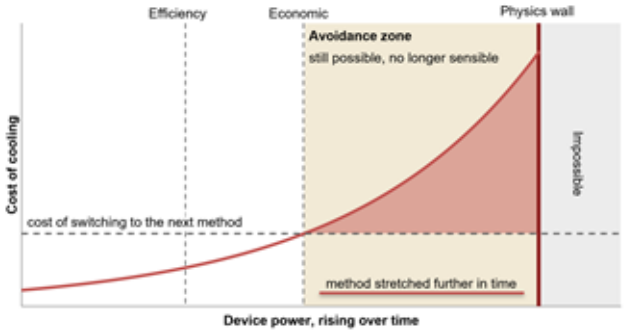
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John Fernandes
Meta

$$\dot{Q} = \dot{m} \cdot c_p \cdot \Delta T$$







Air	1-Phase Cold Plate	2-Phase Cold Plate	Immersion	2-phase immersion	Microfluidics
the moderate and the installed	the hotspot, now	the flux extension	the whole-rack load	strongest thermals, waits on fluids	inside the package

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Project Co-Leads	Sean Sivapalan Nvidia	John Fernandes Meta

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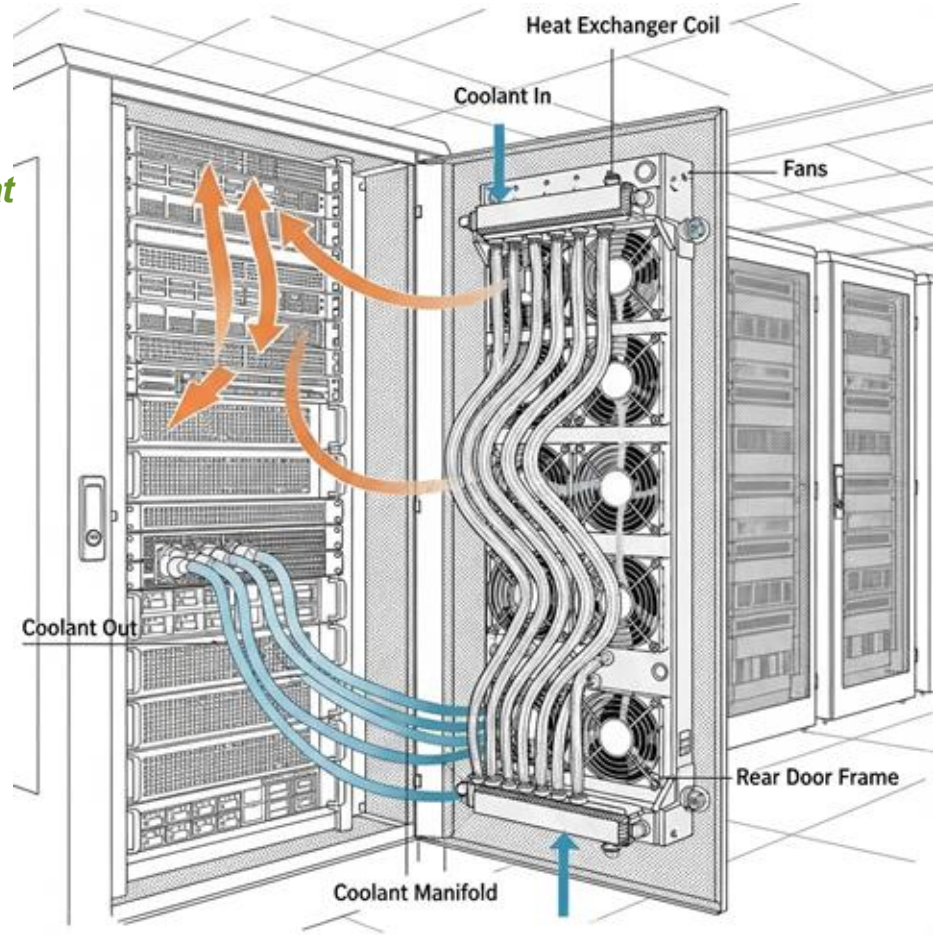
Jessica Gullbrand
(Eaton)

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)

Focuses on the development, integration and standardization of heat exchanger-based solutions



Door Heat Exchanger

Jabari George
(JM Gross Eng)

Martin Doerrich
(Rittal)

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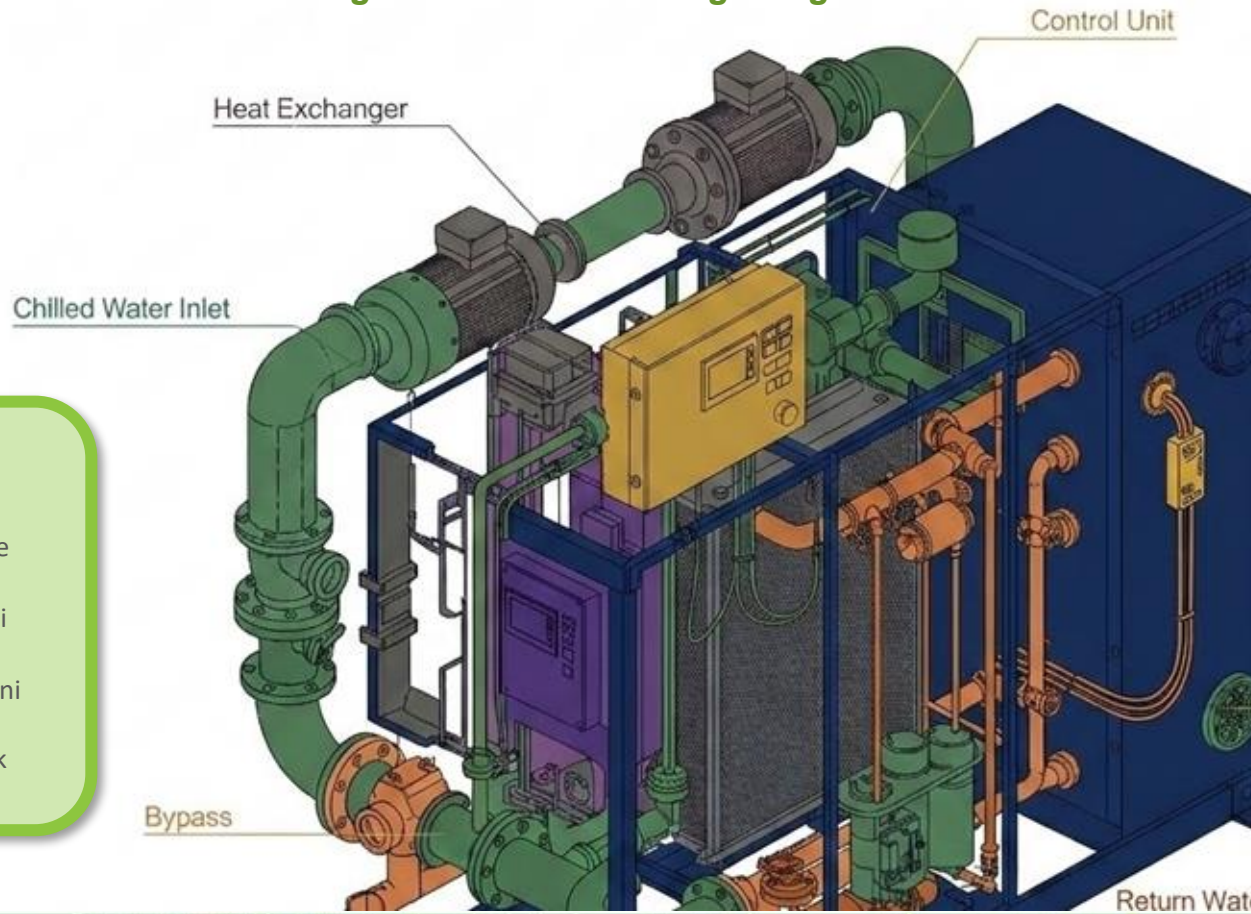
Jessica Gullbrand
(Eaton)

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)

Develop best practices, collaboration documents and common guidelines facilitating integration of CDUs



CDU

John Menoche

Pardeep Shahi

Hussam Kabbani

Brian Burbank

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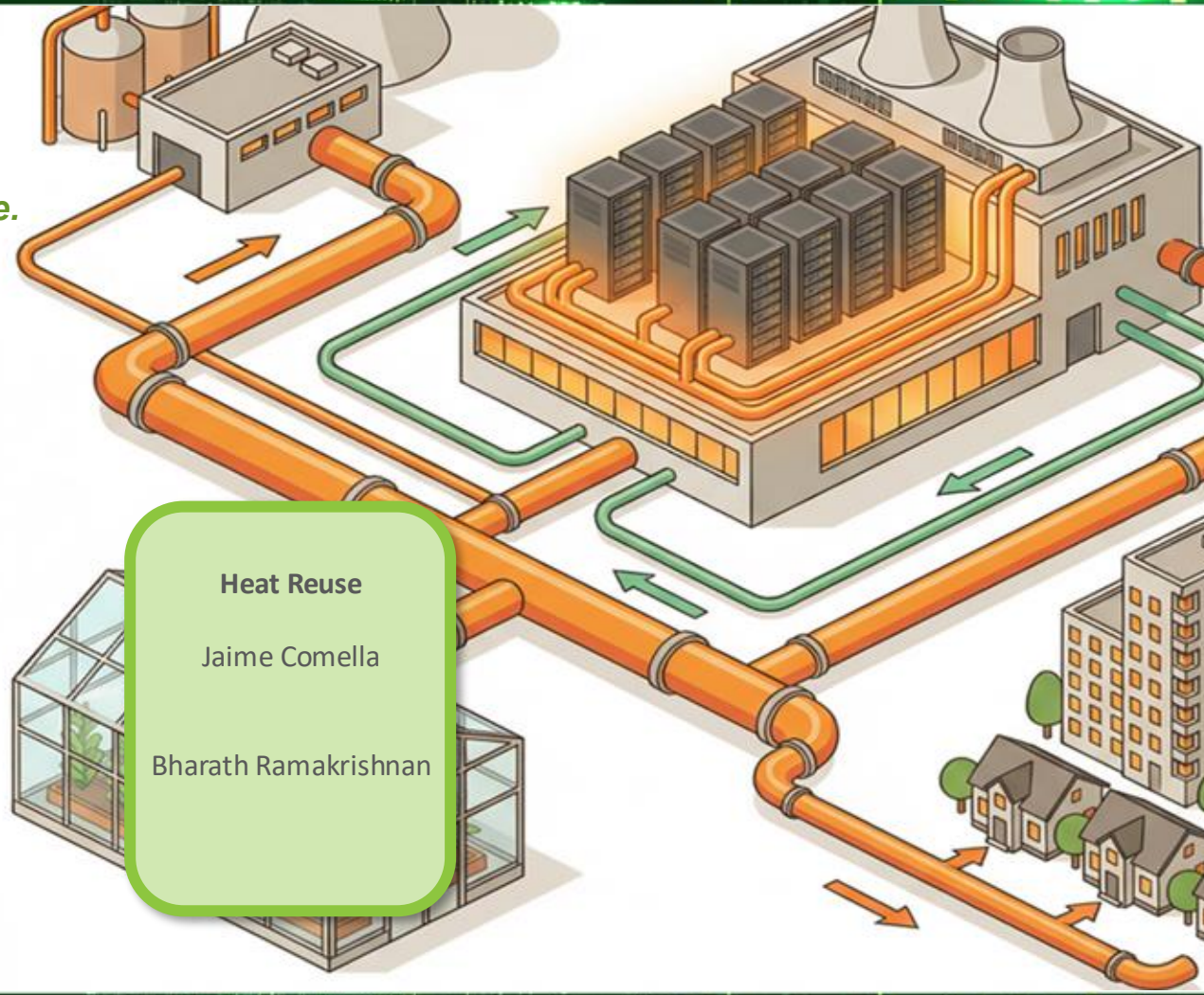
Jessica Gullbrand
(Eaton)

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)

Explores challenges and opportunities with heat reuse. Work is focused/organized across reference designs, economics & policies



Heat Reuse

Jaime Comella

Bharath Ramakrishnan

Cooling Environments Project Structure

Steering Committee Representative

Steve Mills
Meta

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Nvidia

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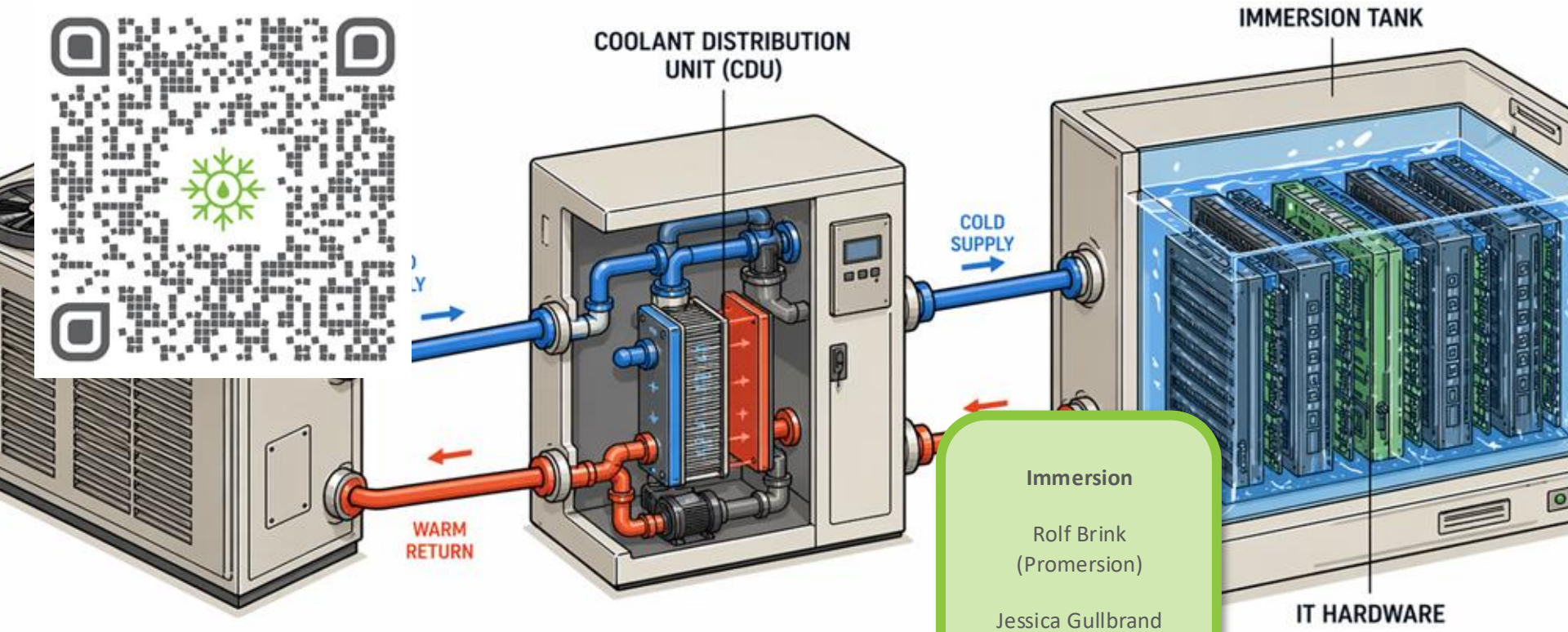
Rolf Brink
(Promersion)

Jessica Gullbrand
(Eaton)

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)



Immersion

Rolf Brink
(Promersion)

Jessica Gullbrand
(Eaton)

Advancing development, adoption, and standardization of immersion cooling technologies

Cooling Environments Project Structure

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Meta

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Nvidia

John Fernandes
Meta

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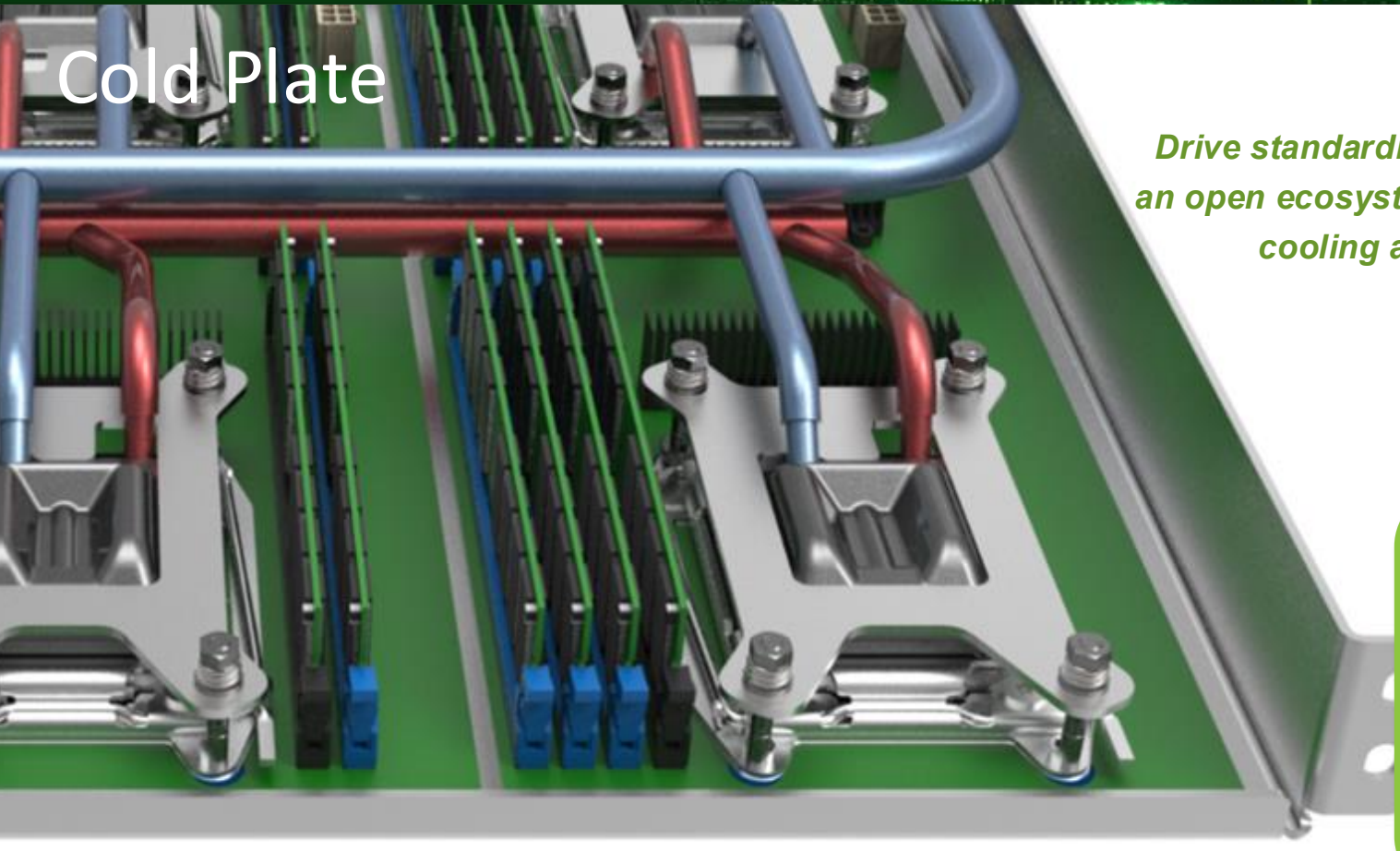
Jessica Gullbrand
(Eaton)

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)

Cold Plate



Drive standardization and enable an open ecosystem for direct liquid cooling applications

Cold Plate

Berhanu Wondimu
(Intel)

Emily Clark
(Dell)

Got Questions?

OCP Liquid Bazaar Sponsored by ExxonMobil

Wed, October 14, 5:15pm - 6:00pm | SJCC - Concourse Level - 220C

“Everything you always wanted to know about liquid cooling, but were afraid to ask”

Moderated by Rolf Brink and Steve Mills

Facilitated by the full Cooling Environments Leadership Team





Platinum sponsors: Johnson Controls and Airsys

Gold sponsors: Shell, Unicom Engineering, Castrol and Shield by Guardian.

Silver sponsors: Viega, Baltimore Aircoil Company, Kyzen, fluidiQ by Magnalytix, ExxonMobil, GRC, Iceotope, Promersion, Alcatex, Avayla, Midas and AIM.

Community partners: Nomad Futurist, Electronics Cooling, GICA and the AI Supply Chain Alliance.

Call to Action

Let’s get Involved!!!




Thank
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October 12-15, 2026
San Jose, California

COOLING
ENVIRONMENTS

Cooling Technologies Testing Workstream & Data Center Cooling Collaborative (DC Cool)

Donghun Kim

Lawrence Berkeley National Laboratory

Cosimo Pecchioli

Singleton Group

OCP BRONZE™ Member

The Challenges

dns

Lack of commercial-scale testbeds

- Many advanced cooling technologies exist, but customers are **hesitant to adopt** them without evidence of reliable performance under realistic data-center conditions

rule

Lack of common testing protocols

- Difficult to compare technologies fairly.
- **Real-world production performance can differ** from reported test performance.

verified

Lack of shareable data and need for neutral third-party validation

- Field data are often proprietary, and generated under different conditions, making meaningful comparison difficult.
- **Neutral third-party** testing is needed to build confidence

Purpose of Workstream

Our Mission

The Workstream aims to accelerate the development and **adoption** of data center cooling technologies through focused subgroups

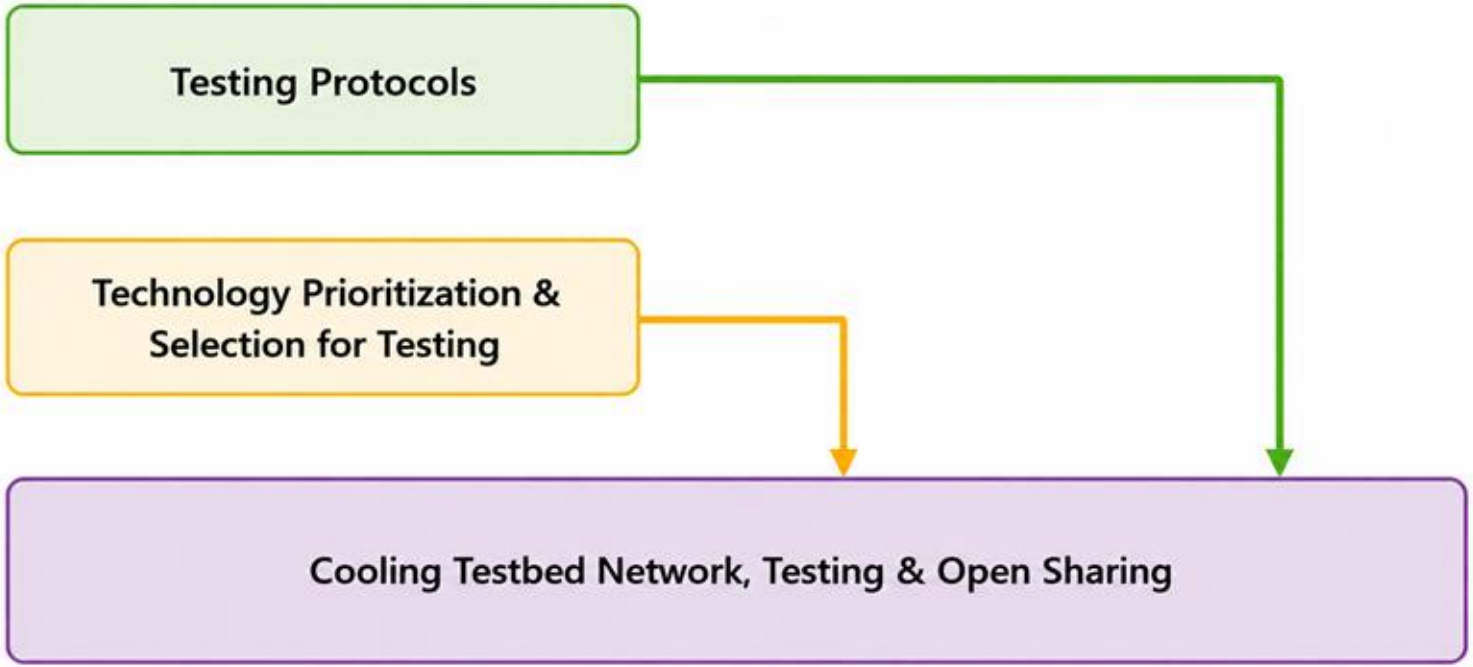
01 **Testing Protocols:** Developing protocols to **help adopters** make more confident decisions, and **help facilities** develop more standardized M&V plans, with the potential for comparative data.

02 **Technology Selection for Testing:** Identifying and prioritizing high-impact cooling technologies for testing

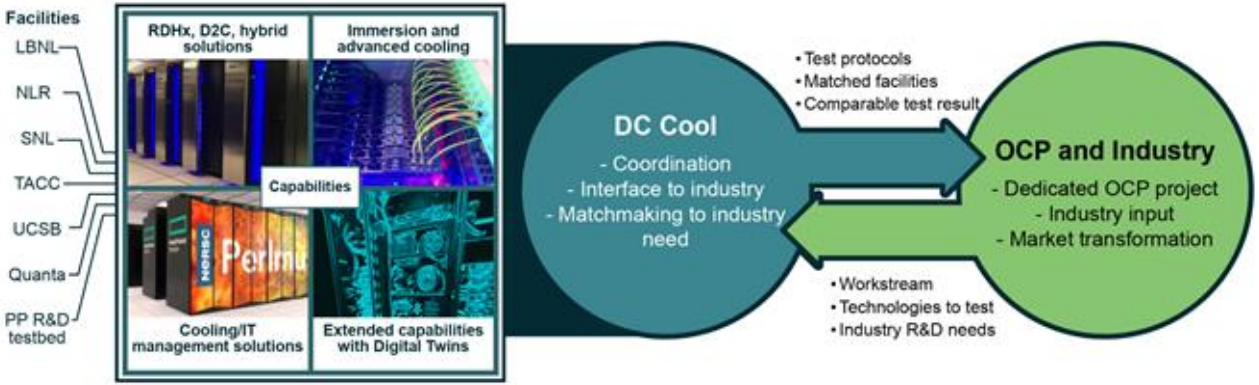
03 **Cooling Testbed Network, Testing and Open Sharing:**

- Maintaining a network capable of commercial-scale evaluation.
- Performing actual tests of selected technologies using standardized testing protocols across a network of participating testbeds; and promoting the sharing of test results, guidance, and best practices, where practicable.

Subgroups and Overall Workflow



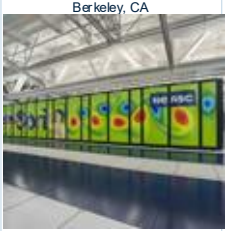
Data Center Cooling Collaborative (DC Cool): Building Momentum for the Workstream



Current Partners & Ecosystem

Berkeley Lab · National Laboratory of the Rockies · Sandia National Laboratories · TACC · OPEN Compute Project · bp Castrol · sabey Data Centers · Schneider Electric · Supermicro · METAseismic · Promersion · Google

NERSC / LBNL
Berkeley, CA



B50B-1275 / LBNL
Berkeley, CA



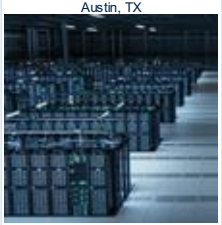
SNL
Albuquerque, NM



NLR ESIF
Golden, CO



TACC
Austin, TX



PP R&D testbed
TBD



Outcomes

- Real-world operational environments
- Industry-consensus testing protocols
- Broad technology coverage
- Design and control optimization using digital twins
- Accelerated market transformation

Cooling Tech Prioritization | **Testing Protocols** | **Best Practice Guidelines for Retrofit** | **Digital Twins** | **Workforce Development and Dissemination**

Call to Action

- Help shape WS priorities and activities
- technical expertise or advisory support through the WS meetings
- Provide access to a test facility
- Participate in a Public-Private R&D Testbed
- Provide cooling equipment, components, or prototypes for testing

Where to find additional information (URL links)

- Cooling Technologies Testing Workstream mailing list:

TBD

- DC Cool website

TBD




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COOLING
ENVIRONMENTS

Industry Liaison Team Update

Kelley Mullick, *Avayla Consulting Group*
Maryam Mohagheghi, *HF Sinclair*

OCP PLATINUM Member

Overview

- Purpose: facilitate two way communication between external standards organizations and OCP to help drive the standards that accelerate adoption of liquid cooling technology

Where We Engage



Monthly call: every 3rd Thursday, 8 to 9am PT | 10 to 11am CT | 17:00 to 18:00 CET

Industry Liaison Team Update: August 2026

ASTM D27

Two standards now in development, initiated following the June Committee Week in Dallas: a Minimum Requirements standard for single phase fluids and a Compatibility standard.

ASHRAE

Matt Koukl, OCP Liaison to ASHRAE TC9.9, will share the ASHRAE update following the ILT update.

White Papers

“Certified AI Datacenters” continues development. NEW: “Thermal Boundaries in AI Infrastructure” has published, setting out a cooling roadmap that spans technologies.

ASTM D27.04 Progress Update

ILT WORKSTREAM UPDATE

Chair & Scope

Griffin Burk (Ergon) leads dielectric liquid standards for immersion cooling

Dallas Meeting

June 3 to 4: strong engagement; approval now underway via D27.15

	Producer	User	Consumer	General Interest	Unclassified	Total
Official Voting Member	14	4	0	18	0	36
Non Official Voting Member	3	0	0	1	0	4

D27.04 voting membership breakdown (Aug 2026)

2 Standards in Development

Minimum Requirements (Burk) & Compatibility (Lance Lewand, Doble)

Join D27.04

Voting membership required; visit astm.org

White Paper Update: Certified AI Datacenters

A Framework for Trusted, Scalable, and Sustainable AI Infrastructure

ILT WORKSTREAM UPDATE



Scope: North America First

Establishing baseline commissioning standards before broader rollout



Technical Committee Alignment

Coordinated development across OCP technical committees



Industry Leading Standards

A framework for trusted, scalable, sustainable AI infrastructure

Scan to access the document

The Bill Is Real. The Discount Is Readiness.

WHITE PAPER SPOTLIGHT | THERMAL BOUNDARIES IN AI INFRASTRUCTURE

Paying the Preparation Debt Early

Industry liaison: collaboration with ASHRAE, ASTM and the standards ecosystem

> *Preparation debt is an ecosystem's unbuilt readiness for a deferred transition — incurred by no one in particular, due all at once when the change is forced.*

Thermal Boundaries in AI Infrastructure, section 5.7

Thermal Boundaries in AI Infrastructure, an OCP white paper from Promersion, Oracle, Avayla and Meta. CC BY-SA 4.0.

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Three Bodies, One Thermal Path

WHITE PAPER SPOTLIGHT | THERMAL BOUNDARIES IN AI INFRASTRUCTURE

ASHRAE

Facility envelopes, water quality and datacom guidance: the rejection side of the thermal path, on the building's clock.

ASTM

Test methods and materials compatibility for working fluids: reference chemistry a supply chain can design against.

OCP Cooling Environments

Interfaces, telemetry and shared vocabulary at the seam, where the IT and facility envelopes meet.

Each body owns a different segment of the thermal path.
Liaison is what makes them one path.

Scan to access the document

The Liaison Pipeline

WHITE PAPER SPOTLIGHT | THERMAL BOUNDARIES IN AI INFRASTRUCTURE



Terminology

One harmonized glossary across CE and the bodies: one vocabulary for the same physics.



Fluids

Fluid test standards and compatibility methods, retiring the no reference fluid string.



Interfaces

Interface definitions from IT to facility: one server, many facilities; one facility, many servers.



Commissioning

Bring up procedures for sites running multiple cooling technologies: the readiness of the site itself.

Every item retires a string the adoption law says the ecosystem will otherwise defer around.

Scan to access the document

Soften the Landing

WHITE PAPER SPOTLIGHT | THERMAL BOUNDARIES IN AI INFRASTRUCTURE

Readiness Converts the Transition

A standard ready before the forcing function arrives turns a scramble into a deployment. The cost of readying early is small next to the bill of arriving unprepared.

The Ask

Bring your requirements to the liaison table: fluids data, interface needs, commissioning experience. The bodies move at the pace of contributed evidence.

Read it. Break it. Feed it data.

Thermal Boundaries in AI Infrastructure, an OCP white paper from Promersion, Oracle, Avayla and Meta. CC BY-SA 4.0.

Scan to access the document

Call to Action

Industry Liaison Opportunities

We are seeking an IEC TC 108 industry liaison representative to OCP. If you work with an organization our team does not yet cover, we would love to hear from you.

Join Our Team

Reach out to Kelley Mullick (Kelley.Mullick@ocproject.net) or Maryam Mohagheghi (Maryam.Mohagheghi@ocproject.net).

Stay Updated

Monthly call every 3rd Thursday: 8 to 9am PT | 10 to 11am CT | 17:00 to 18:00 CET. Scan the QR code to join the ILT mailing list.

Working Documents

The QR code also links to the working documents for Certified AI Datacenters and Thermal Boundaries in AI Infrastructure. Read them, comment and contribute.

Mailing list and working documents



COOLING
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Cooling Environments- ASHRAE TC9.9 & EE HPC WG collaboration updates

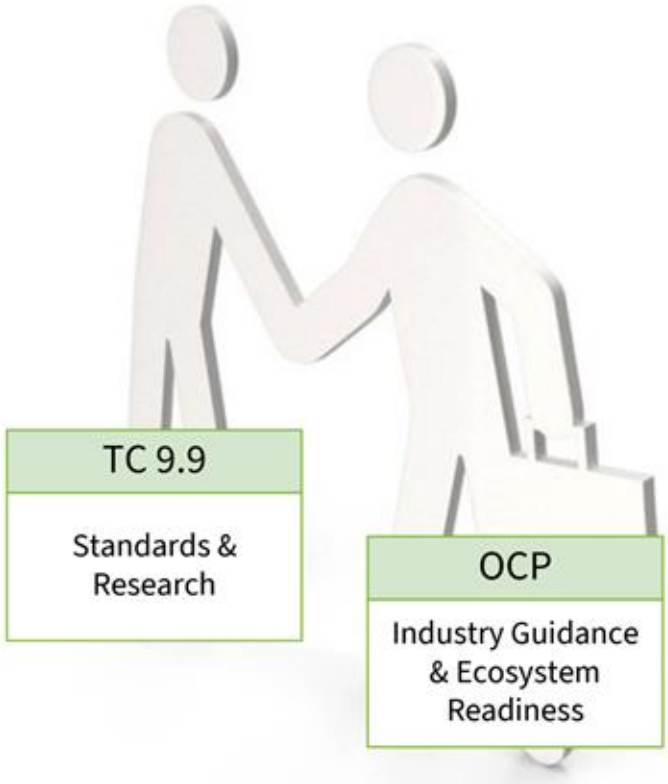
Matt Koukl

OCP Liaison, ASHRAE TC9.9

Affiliated Engineers, Inc.

Why Collaborate?

- Generate wider knowledge base
  - Leverage unique strengths
  - Coordination of different voices in the value chain
  - Engagement from silicon to ambient
- Educate
  - Avoid duplication of work
- Collaborate
  - Shared/aligned roadmap
- Communicate
  - Published work; publicizing



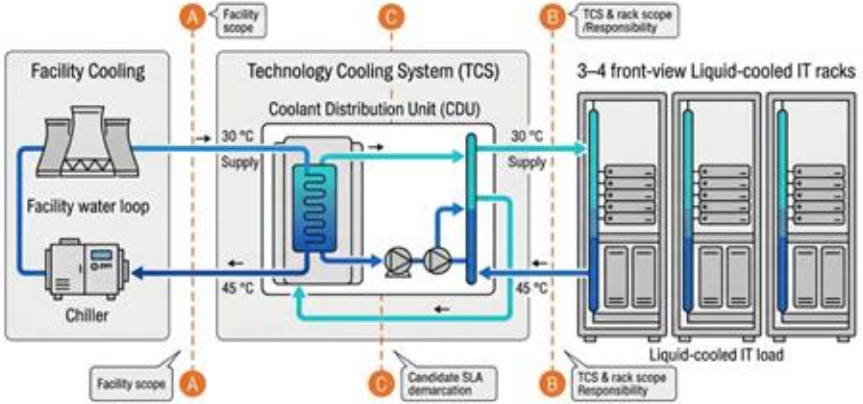
Activity Since Inception

- MoU signed between ASHRAE & OCP
- ASHRAE TC 9.9 OCP Liaison role established
- Aligned activities
  - Flow velocity & erosion workstream (Cold plate SPJ)
  - Guidelines for pre-commission preparation of TCS row manifolds
  - Fluids workstream (Cold plate SPJ)
  - Initiation of Cooling Environments CDU SPJ
- Underway
  - PG25 mixing studies
  - RTAR for fluid velocity approved
  - Pumped two-phase cooling white papers
  - FWS pre-commissioning guidelines (w/ EE HPC WG & OCP DCF PJ)
- Panel session at 2026 ASHRAE annual conference



Soliciting Participation & Feedback

- *Upcoming area*: Liquid Cooling SLA Demarcation in Data Centers
  - Traditional (air cooled) SLA demarcation is no longer fit for purpose
  - WP identifies core tensions in defining a new SLA boundary between colocation providers and their customers in liquid-cooled environments
  - Examines practical tradeoffs between different demarcation models
  - Kickoff meetings near the end of the year



Call to Action

Soliciting feedback & participation on relevant & critical areas

Please reach out to Matt Koukl - matt.koukl@ocproject.net or the CE project co-leads and SC sponsor




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COOLING ENVIRONMENTS

Accelerating Liquid Cooling Interoperability: AI-Driven Redfish Schema Development- Mock-Ups- and OCP Profile Validation

Mick Jones

Principal Architect - Vertiv

OCP PLATINUM Member

Purpose of Workstream

- Define a unified schema under DMTF Redfish for management of all OCP Liquid Cooling Equipment:
  - CDUs, Immersion; Tank and Enclosed Chassis
  - Cold Plate; BMCs, aggregation and control loops, Rear Door HX
  - Facilities; FWS, valves, inline sensors, Chillers#, etc
  - Heat Re-use; Tanks/Reservoirs, Heat Exchangers, Pumps, etc
- Create mockup (examples) of the expected output from the compliant management interfaces and management systems
- Contribution schema and mockups to Redfish Forum for inclusion in Redfish release
- Create/extend OCP Interoperability Profiles

The Challenge

- Growth in device types, schemas and profiles creates scaling challenges
  - Backlog of 10+ mockups and Interop profiles
- Manual schema authoring
- Manual mockups
- Manual validation
- Limited test coverage versus schema, protocol and profile validation

Where AI fits

AI assists in:

- Schema development
- Resource modelling
- Mockup generation
- Profile validation
- Edge-case identification
- Consistency checking/maintenance

Outcome

✅ Faster development

✅ Better quality

✅ Higher coverage

AI-assisted lifecycle: (replace with diagram later)

Requirements → Schema Generation → Mockups → Profile Validation → Interoperability

AI for Mockup Generation

Traditional

- Build JSON payloads manually
- Create example resources individually
- Time-consuming updates

AI Driven

- Generate realistic resource trees
- Produce example JSON payloads
- Create usage guides and diagrams
- Rapidly iterate designs

Outcome

Faster stakeholder review and standards refinement.

Example Coverage

- CDU systems
- Immersion systems
- Sensors
- Pumps
- Valves
- Facility integrations

AI For Interoperability Validation

AI Validation Capabilities

- ✅ Profile conformance checking
- ✅ Missing property detection
- ✅ Requirement traceability
- ✅ Cross-schema consistency checks
- ✅ Edge-case generation
- ✅ Coverage analysis

Example

Profile Requirement:

- Property must exist
- Value type must comply
- Resource relationships must be valid

AI automatically reviews thousands of combinations.

Accelerating Hardware Management for LC

Key Messages

1. HMLC defines standard Redfish management models for liquid cooling.
2. AI accelerates schema development and maintenance.
3. AI rapidly generates realistic mockups.
4. AI improves profile creation, validation and documentation.
5. Combined workflows reduce effort and increase consistency.

Closing Statement

AI enables the OCP community to move from schema design to validated interoperability faster, improving standardization for next-generation liquid-cooled infrastructure.

Call to Action

Help create standard workflows to accelerate mockup and interoperability profile creation for HMLC

Attend the HW Liquid Cooling WS meetings (the more perspectives, the better)

- Provide input on manageability requirements (encapsulated in profile)
- Provide expertise for iterations

Where to find additional information (URL links)

-Hardware Management for Liquid Cooling mailing list:

- https://ocp-all.groups.io/g/OCP-HWmgt-Liquid-Cooling

-OCP Interop Profiles:

https://github.com/opencomputeproject/HWMgmt-OCP-Profiles

SCAN TO JOIN




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COOLING ENVIRONMENTS

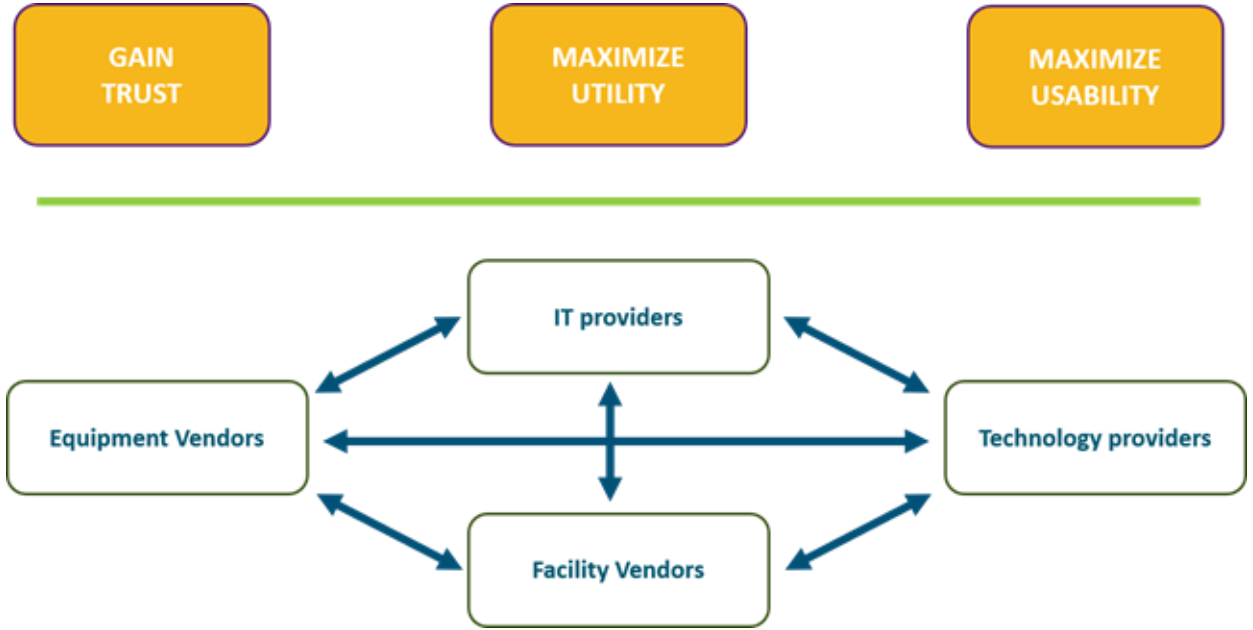
Community Update: Total Cost of Ownership Workstream

Dr Andy E Young, *Asperitas*
Eduard Roytman, *Microsoft*

OCP PLATINUM Member

TCO model: Principles

- Openness
- Flexibility
- Trust



TCO model: Openness

Open Model: build entirely in Excel spreadsheet, no macros, nothing hidden, fully trackable with a detailed User's Guide and Quick start sheet

Open Data: power, cooling and IT equipment stored in open libraries and organized in configurations

Open Definitions: modelling approach and assumptions are fully documented in a Reference Manual aligning to the OCP glossary.

Other Model exist, this one is uniquely

- Open-source (no costs, paywall, proprietary data)
- Open-ended (extendable without fundamental changes)
- Comprehensive (bottom-up & top-down)

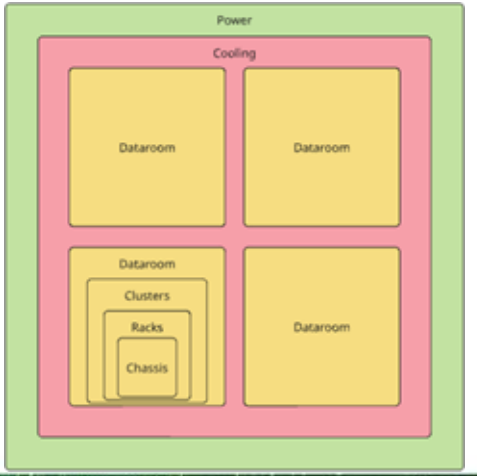
TCO model: Flexibility

- Capable to represent a wide range of technologies, for power, cooling and IT equipment (data)
- Supporting all forms of air- and liquid-cooling
- Hybrid combinations
- Range of geographic locations

IT Operations		Paris	Barcelona
Power setpoint (position)		L1	L1
Power capacity utilization		100%	100%
Dataroom 1	type	DLC+RDHX	DLC+RDHX
	power setpoint	50,000kW	50,000kW
Dataroom 2	type	None	None
	power setpoint	0kW	0kW
Dataroom 3	type	None	None
	power setpoint	0kW	0kW
Dataroom 4	type	None	None
	power setpoint	0kW	0kW

Facilites Operations		Paris	Barcelona
Power configurations	Critical	SE RD Crit	SE RD Crit
	Mechanical	SE RD Mech	SE RD Mech
	Redundancy	2N	2N
Cooling configurations	Air	Default Air	Default Air
	Liquid	Default Liquid	Default Liquid
	Redundancy	N+1	N+1
Environment Zone	Temperature	Mixed	Warm
	Humidity	Humid	Dry



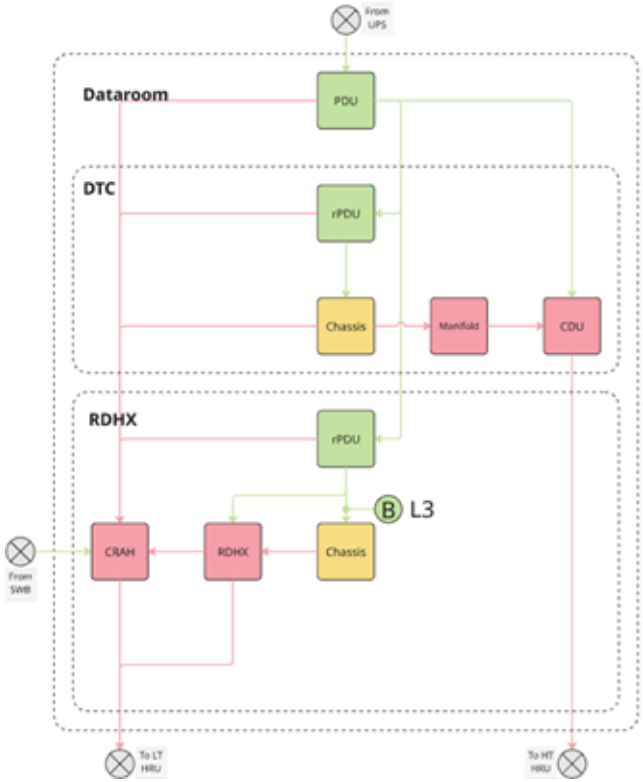


Finanical Operations	Paris	Barcelona
Metering position	L3	L3
Utility electricity unit cost	0.1€/kWh	0.1€/kWh
Core and shell unit cost	€1800/m2	€1800/m2
Fit-out unit cost	€1800/m2	€1800/m2
Water unit cost	1.00€/m3	1.00€/m3
Value of heat recovery	0.00€/kWh	0.00€/kWh
Core and shell maintenance	3%	3%
Equipment maintenance	3%	3%
electricity CO2 impact	369g/kWh	369g/kWh
electricity water impact	1.8l/kWh	1.8l/kWh
Facility, Power and Cooling lifes	18 yr	18 yr
IT equipment lifespan	6 yr	6 yr
Discount rate for NPV	5%	5%
CAPEX Financing rate	6%	6%
CAPEX % Financed	70%	70%
CAPEX Years Financed	15 yr	15 yr
Annual hours	8760	8760

TCO model: Flexibility

- CDU placement in Cluster or in Rack
- Air-cooling at Dataroom, Cluster and Rack levels
- Variable % for air- and liquid-side heat rejection
- Hybrid configurations

- Air-Cooling (100% air-side load)
  1. Perimeter CRAH
  2. In-row CRAH
  3. DTC with L2A CDU (side-car)
  4. Open (passive) RDHX
  5. DTC with L2L CDU
  6. Closed (active) RDHX
  7. Immersion
- Liquid-Cooling (<5% air-side load)



TCO model: Access and support

- OCP tool v1.31 released
- New features
  - Input: More top-down driven
  - Model: Improved finance calculations
  - Model: CDU placement now in cluster
  - Output: Breakdown overhead and losses
- Enabling new insights

Download the tool





Call To Action

Try the tool -

- Download from OCP Marketplace or from Drive
- Apply to your Datacenter scenario, Triangulate with Other Models
- Cooling equipment vendors - add your products to the tool lib

Thinking of making this tool even better?

- Get involved - reach out to OCP CE Solutions leadership




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COOLING ENVIRONMENTS



Experimental Framework for Determining Coolant Flow Velocity Limits in Liquid-Cooled AI Factory TCS Loops

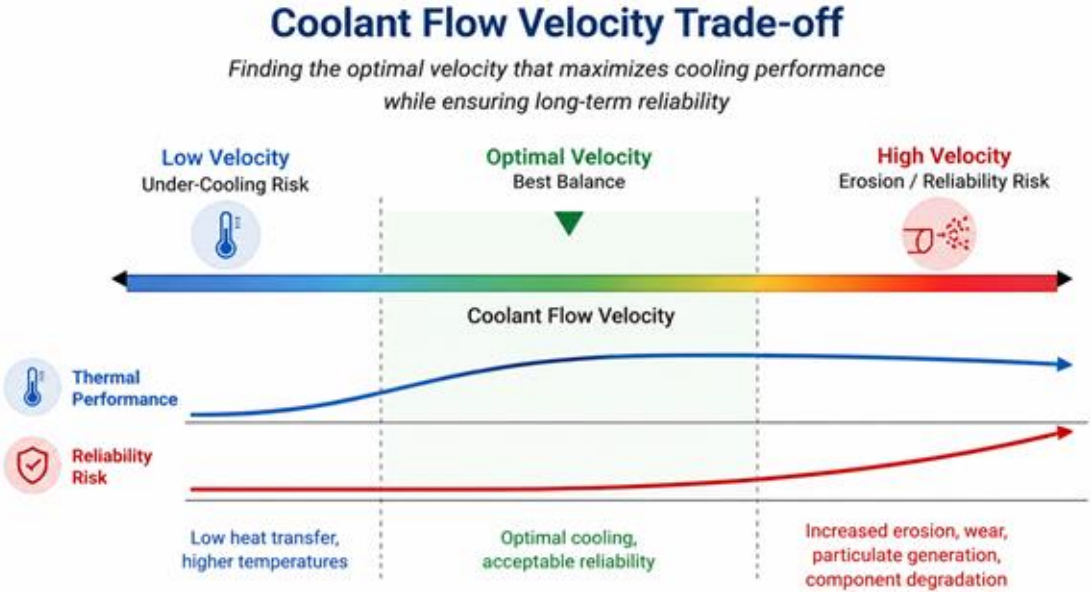
Qusai Soud, *Nvidia*
Mark Steinke, *Nvidia*

Problem Statement

The liquid cooling industry lacks standardized, evidence-based coolant flow velocity limits, creating uncertainty in balancing thermal performance and long-term reliability.

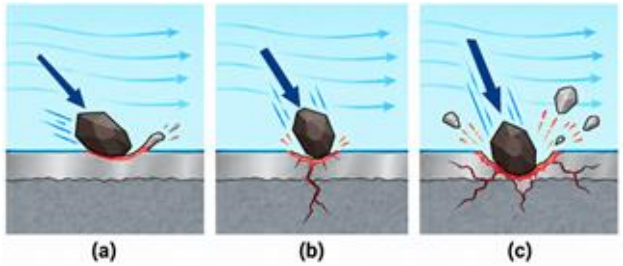
Why It Matters

- Higher coolant velocity improves thermal performance.
- Excessive velocity can accelerate erosion, wear, and particulate generation.
- Conservative limits increase pumping power or restrict cooling performance.
- Unvalidated limits increase reliability risk.

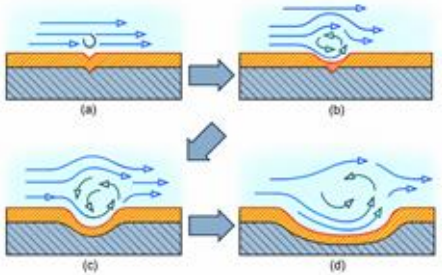


Sources of Erosion

Particle-induced erosion progression:



Flow-Induced erosion progression:



Erosion Type	Impact	Key Cause	Common Conditions
Liquid-Particle Erosion	High	Solid particle impacts	Slurries, contaminated coolants
Single-Phase Erosion	High	High velocity, turbulence	Bends, valves, reducers
Cavitation Erosion	Moderate	Bubble collapse	Pumps, low-pressure regions
Erosion-Corrosion	Low	Mechanical + chemical degradation	High velocity with corrosive fluids
Flashing-Induced Erosion	N/A	Sudden vaporization	Refrigerant expansion, high ΔP areas

Sources of Particle-induced Erosion

TCS Contamination & Erosion

- TCS loops are inherently contaminated from manufacturing and installation.
- Fabrication debris: Inadequate controls for welding, cutting, deburring, and passivation can leave burrs, weld slag, and other debris.
- High-velocity risk: Higher coolant velocities can accelerate entrained particles, increasing erosion risk.



Rough Cut Edges



Weld Oxidation/Heat tint



Metallic Debris



Lack of Shielding/Passivation

Testing for Erosion and Velocity Limits in TCS Loops

01	Identifying high velocity components	• Velocity through Internal flow paths • Component material • Shape and surface finishing
02	Experimental setup design and test matrix	• Oversizing setup piping • FNM and pump sizing • Preserving geometry, pressure, and temperature
03	Test duration	• Acceleration factors • Fluid sampling frequency
04	Fluid health and suspended particles control	• Fluid health parameters monitoring (Ph, Conductivity, Turbidity..etc) • Tracking the type, count, and size of suspended particles

Analysis and Inspection

Non-Destructive Test (NDT) Methods

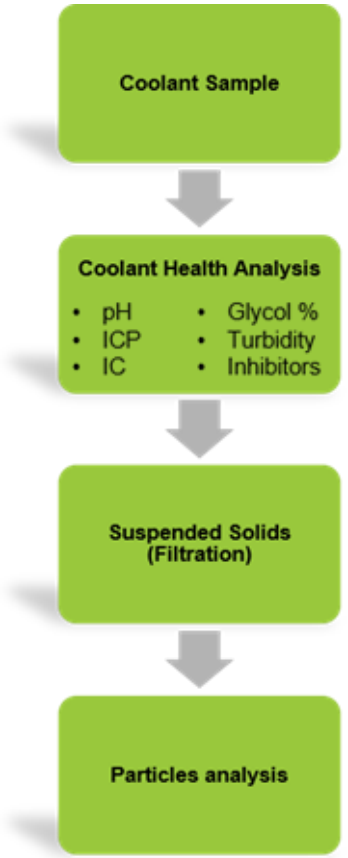
Method	Smallest Detectable Defect	Notes
Borescope / Videoscope	100–200 µm	Best for visual morphology; limited depth assessment; works around bends.
Ultrasonic Thickness (UT) Gauging	≥ 0.5 mm	Single-point or small-area measurement; low accuracy.
Phased Array Ultrasonic Testing (PAUT)	≥ 0.2–0.5 mm	Produces 2D/3D wall maps; good for pitting and erosion depth quantification.
Laser Profilometry / Structured Light (Mini)	≥ 10–50 µm	Requires internal dry conditions, may need borescope-based mounting.

Destructive Test Methods

Method	Smallest Detectable Defect	Notes
Optical Microscopy	~1–2 µm	Requires polished cross-section; useful for depth and shape measurement of pits or scratches.
Scanning Electron Microscopy (SEM)	<0.1 µm	Excellent for analyzing erosion mechanisms; can detect micro-cracks from cavitation or particulate impingement.

Fluid Analysis

Coolant Parameters Tests	**ICP:** Measures dissolved metals in the coolant. **IC (Ion Chromatography):** Measures ionic contaminants. **PH:** Measures coolant acidity/basicity. **Reserve Alkalinity:** Measures remaining buffering capacity. **Turbidity:** Measures how cloudy the coolant is. **AZOLE:** Measures azole inhibitor concentration. **Glycol RI (refractive index):** Uses refractive index to determine glycol concentration.
Suspended Solids (Filtered) + IR	Characterization of solid deposits using infrared (IR) analysis for any solids found in the fluid. If insufficient solids are collected on the filter during Total Suspended Solids (TSS) analysis, IR analysis may not be performed.
Particles SEM, EDS, Particle Count	SEM characterizes particle morphology and features, while EDS identifies the elemental composition of the filtered debris.

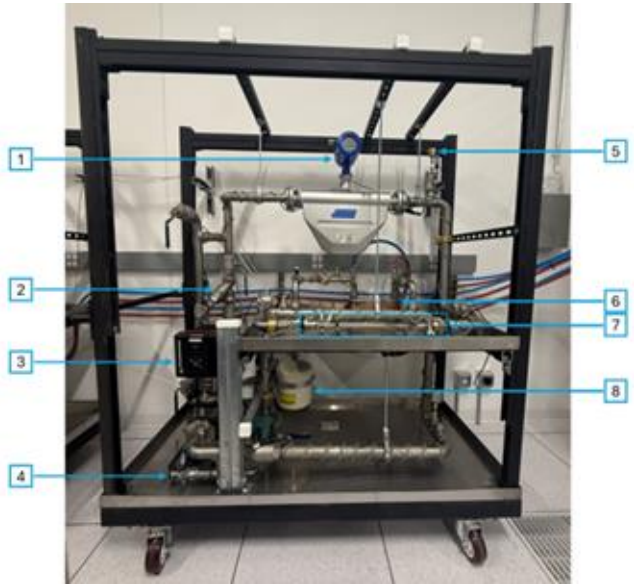


Experimental Setup

- Closed-loop liquid cooling test bench with powerful circulation pump.
- PG25 coolant circulated under controlled conditions
- Multiple high-risk components tested simultaneously
- Continuous monitoring of flow rate, pressure, temperature, and pump operation
- Periodic fluid sampling for chemistry and particulate analysis
- Pre and post-test CT scans and microscopy to quantify material degradation



Liquid-to-Air CDU providing cooling for temperature control



Component-level test setup:

1. Coriolis flow meter
2. Y-strainer (can either be inline or side-stream)
3. Centrifugal pump
4. Drain port
5. Air vent
6. Heat exchanger
7. Test specimen
8. Expansion tank

Emulating Suspended Particles

Particle Type	Examples	Typical Hardness (Mohs Scale)	Density (kg/m3)	Erosive Impact Level
Metallic Oxides	Copper Oxide, Aluminum Oxide	**3.5 – 9.0** (Very Hard)	4,000 – 6,000	**Highest**
Metal Shavings	SS, Copper, Iron, Brass	**2.75 – 6.5**	2,700 – 8,900 (Very dense)	**Highest**
Mineral Scales	Calcium Carbonate (CaCO3), Silica	**3.0 – 7.0** (Hard/Brittle)	2,500 – 2,700	**Moderate**
Polymeric Debris	EPDM fragments, gasket flakes	**< 1.0** (Soft/Elastic)	900 – 1,200	**Low**
Biological Matter	Biofilms, bacteria colonies	**< 0.1** (Soft/Sticky)	1,000 – 1,100	**Negligible**

Table 1 Water Quality Guidelines for the FWS and TCS

Parameter	FWS (Table 5.3, ASHRAE 2014)	TCS (Table 6.2, ASHRAE 2014)
pH	7 to 9	8.0 to 9.5
Corrosion inhibitor(s)	Required	Required
Biocide	—	Required
Sulfide	<10 ppm	<1 ppm
Sulfate	<100 ppm	<10 ppm
Chloride	<50 ppm	<5 ppm
Bacteria	<1000 CFUs/mL	<100 CFUs/mL
Total hardness (as $CaCO_3$)	<200 ppm	<20 ppm
Conductivity	—	0.2 to 20 micromho/cm
Total suspended solids	—	<3 ppm
Residue after evaporation	<500 ppm	<50 ppm
Turbidity	<20 NTU (Nephelometric)	<20 NTU (Nephelometric)

ASHRAE Technical Committee 9.9. (2019). Water-cooled servers: Common designs, components, and processes. ASHRAE.

Call to Action

Help define evidence-based coolant velocity limits for TCS loops

- Share component failure / erosion data
- Contribute candidate components for testing
- Share material, coating, and field-condition data
- Participate in defining test methods and acceptance criteria

Continue the discussion through the OCP Cooling Environments community




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